Future Engineering Journal

Volume 4 | Issue 2 Article 2

September 2024

Utilizing Deep Learning in Smart Glass System to Assist the Blind and Visually Impaired

Asmaa A. Hekal
Modern University for Technology and Information, asmaa.hekal@eng.mti.edu.eg

Mohamed S. Sharaf
Modern University for Technology and Information, mohamed.sharaf@eng.mti.edu.eg

Ahmed A. Sayed
Modern University for Technology and Information, ahmed.87046@eng.mti.edu.eg

Ibrahim R. Abdelrahman
Modern University for Technology and Information, ibrahim.86607@eng.mti.edu.eg

Ahmed A. Salem
Modern University for Technology and Information, ahmed.88100@eng.mti.edu.eg

See next page for additional authors

Utilizing Deep Learning in Smart Glass System to Assist the Blind and Visually Impaired

Authors

Asmaa A. Hekal, Mohamed S. Sharaf, Ahmed A. Sayed, Ibrahim R. Abdelrahman, Ahmed A. Salem, Ahmed M. Elhussieny, Saeed Y. Kouta, and Eman S. Abass

FUTURE ENGINEERING JOURNAL 4 (2024) 2314-7237

Future Engineering Journal

Journal homepage: https://digitalcommons.aaru.edu.jo/fej/

Utilizing Deep Learning in Smart Glass System to Assist the Blind and Visually Impaired.

Asmaa A. Hekal [a], Mohamed S. Sharaf [a], Ahmed A. Sayed [a], Ibrahim R. Abdelrahman [a], Ahmed A. Salem [a], Ahmed M. Elhussieny [a], Saeed Y. Kouta [a], and Eman S. Abass [a]*

[a] *Electrical Engineering Department, Faculty of Engineering, Modern University for Technology and Information (MTI University), Cairo, Egypt*

ARTICLEINFO

Article history:
Received: November 2023
Received in revised form: June 2024
Accepted: August 2024

Keywords:
Smart Glasses
Artificial Intelligence
Blind and Visually Impaired
Deep Learning
Object Detection
Face Recognition

ABSTRACT

This paper presents a groundbreaking assistive technology designed to empower visually impaired individuals in their daily lives. With an estimated global population of 2.2 billion facing visual impairments, addressing the challenges they encounter is of paramount importance. The research introduces a comprehensive electronic device integrating advanced computer vision and deep learning techniques. The system incorporates real-time object detection, robust facial recognition, and precise currency denomination identification. Powered by a Raspberry Pi 4 Model B+ and an ESP32-CAM Development Board, the device offers users unparalleled environmental awareness. Utilizing YOLOv4-tiny for object detection and a hybrid face recognition model combining HaarCascades, Histogram of Oriented Gradients (HOG) features, and deep neural networks, the system ensures accurate object and facial recognition. The integration of a MobileNetV2-based model facilitates Egyptian currency detection. Moreover, the system features seamless speech-to-text and text-to-speech functionalities, enhancing user interaction. The device's hardware components, including ultrasonic sensors and vibration motors, provide haptic feedback, further augmenting the user experience. This paper not only presents a state-of-the-art technological solution but also discusses its future potential. Through continuous refinement, user feedback integration, and exploration of wearable applications, the proposed system holds promise in significantly improving the quality of life for visually impaired individuals, fostering independence, and facilitating greater societal inclusion.

Peer review under responsibility of Faculty of Eng. & Tech., Future University in Egypt.

1. Introduction

In the contemporary landscape of information and communication technology, the lifestyle and autonomous mobility of individuals afflicted by visual impairments represent paramount societal concerns.

As per the World Health Organization, a staggering 2.2 billion individuals globally grapple with vision impairment or blindness, underscoring the urgency and magnitude of this issue [1]. Blind and visually impaired individuals commonly face challenges related to action execution and environmental awareness in their daily lives. Numerous solutions have been proposed to address these challenges, incorporating navigation and object recognition techniques. However, the most prominent navigation methods, including canes, trained guide dogs, and smartphone applications, come with their limitations. For instance, traditional canes prove ineffective in long distances and crowded environments, failing to provide crucial information about hazardous objects or oncoming traffic when crossing roads. Additionally, guide dog training is both laborious and costly, demanding specialized care and attention. While smartphone applications like voice assistance and navigation maps tailored for individuals with visual impairments are advancing, their widespread and effective utilization remains limited.

* *Corresponding author.* Tel.: E-mail address: mohamed.sharaf@eng.mti.edu.eg
Peer review under responsibility of Faculty of Engineering and Technology, Future University in Egypt.

https://digitalcommons.aaru.edu.jo/fej/

Recent advancements in embedded systems and artificial intelligence have revolutionized the field of wearable assistive technologies designed for individuals with visual impairments. These advancements have led to the development and commercial availability of various innovative devices. For example, Daescu and colleagues [2] proposed a system incorporating smart glasses with a face recognition feature. Their approach utilized a server-based deep learning face recognition model within a client-server architecture, optimizing power consumption and computational time. Another notable development comes from Joshi et al. [3], who introduced an assistive device capable of object recognition through a deep learning model. To enhance its functionality, they integrated a distance-measuring sensor, allowing the device to identify barriers during travel between different locations. These examples highlight the continuous progress in creating wearable assistive technologies, leveraging the power of embedded systems and artificial intelligence to improve the lives of visually impaired individuals. In response to the challenges faced by visually impaired individuals, this research endeavors to create an electronic device integrating object and color detection, face recognition, currency detection, and audible feedback. The device is meticulously crafted to empower the independence of individuals with visual impairments, with a primary emphasis on enhancing navigation assistance and heightening awareness of the surrounding environment within this community. The proposed system combines cutting-edge technologies, including embedded systems and artificial intelligence algorithms, to provide real-time assistance to users. For instance, object and color detection functionalities enable users to identify objects and determine their colors, enhancing environmental awareness. Face recognition capability assists in recognizing individuals, while currency detection helps users identify different denominations of currency. Additionally, audible feedback provides users with vital information in a non-intrusive manner, ensuring seamless integration into their daily lives.

1.1. Related Work

In recent years, substantial research endeavors have focused on advancing intelligent systems tailored to facilitate the safe and independent mobility of visually impaired individuals. These innovative assistive technologies harness artificial intelligence methodologies and sophisticated smartphone applications to augment mobility and self-sufficiency. For instance, Awad et al. [4], researchers introduced an intelligent eye Android mobile application engineered to recognize light, colors, diverse objects, and banknotes for visually impaired users. Light detection was achieved through the smartphone's embedded light system. The authors employed the OpenCV library for color differentiation and utilized a locally stored database for object recognition. The development process involved Java coding in Android Studio, integration of the CraftAR toolbox for image recognition, and utilization of CNNdroid, an open-source library facilitating the execution of Convolutional Neural Network (CNN) models on Android devices. Additionally, the application incorporated a built-in voice engine, enabling the auditory presentation of detection results to the user. Bhatlawande et al. [5] introduced a system for detecting physical movements and kitchen activities, specifically designed to aid visually impaired individuals. This system utilized machine learning methodologies, employing computer vision-based scale-invariant feature transform to extract dataset attributes. Through the application of a Random Forest model with optimized hyperparameters, the system attained a notable accuracy level of 0.808, signifying its effectiveness in assisting blind individuals. Mukhiddinov et al. [6] utilized an enhanced iteration of the YOLOv4 deep learning technique to identify the freshness of fruits and vegetables. The system demonstrated the ability to classify five distinct categories of produce, utilizing a diverse dataset collected from multiple sources and under varying lighting conditions. The improved YOLOv4 model achieved a 69.2% Average Precision (AP50) for the test samples. Additionally, a smartphone application was developed for real-time detection of spoiled fruits and vegetables. Assistive technologies frequently encompass a variety of hardware components that integrate embedded circuits and sensors. In a recent proposal elucidated in [7], the authors introduced a pioneering assistive hardware device. This portable apparatus, engineered using Raspberry Pi and TensorFlow object detection API, is designed to augment spatial awareness for individuals with visual impairments. The system, centralized within eyeglasses, serves as the fundamental infrastructure for multiple hardware modules. Users receive real-time auditory feedback via headphones, enabling immediate comprehension of their environment. In this study, the principal processing unit employed was the Raspberry Pi 3B+. A Raspberry Pi camera facilitated the live feed, while the TensorFlow object detection API was utilized for object recognition. Specifically, the detection model incorporated in this system was SSDLite MobileNetV2, complemented by the integration of eSpeak into the auditory feedback network. Despite the promising features exhibited by the aforementioned system, it lacks certain essential capabilities, particularly the ability to verbally describe objects to users. In a Nishajith et al. [8], a smart wearable cap was designed employing Raspberry Pi 3, a NoIR camera, earphones, and a power supply. The device captures images of objects using the NoIR camera, which is subsequently processed by the Google Brain Team utilizing TensorFlow for object recognition. The system successfully identifies a wide array of objects from 90 distinct classes. To convey information about the recognized objects, earphones are utilized, employing a text-to-speech synthesizer called eSpeak to convert the captured video details into speech signals, thus providing auditory descriptions for the user.

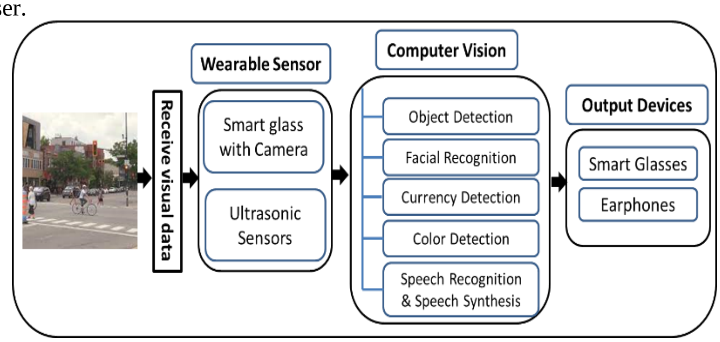


Figure 1. The framework of a system that utilizes smart glass technology delves into the complexities of the computer vision module, covering Object Detection, Facial Recognition, Currency Detection, Color Detection, Speech Recognition, and Speech Synthesis.

Figure 1 shows the architecture of a wearable navigation system integrated into smart glasses.

The main focus of this study delves into the complexities of the computer vision module. The primary contributions of this research are highlighted as follows:

- Development of a fully automated smart glass system tailored for individuals with visual impairments, aiding them in comprehending their surroundings. This system furnishes real-time auditory feedback, offering users information about nearby objects.
- Distinct advantages set this proposed system apart from previous iterations. It harnesses deep learning models in computer vision techniques, surpassing mere object detection. Additionally, it moves beyond basic tracking methods by incorporating four pivotal deep learning models: object detection, facial recognition, currency detection, and Speech-to-Text and Text-to-Speech conversion.

2. Proposed system

In this research, an innovative approach was implemented to enhance the independent navigation of individuals with visual impairments, enabling them to perceive their surroundings autonomously. The key strategy involved the development of a wearable smart glass and a multifunctional system equipped with a mini camera, which captures images and provides object recognition results with voice feedback to users. To facilitate this process, we introduced a client-server architecture, comprising smart glass as the local component and an artificial intelligence server responsible for image processing tasks. The local part includes smart glass. The artificial intelligence server receives images from the local device, processes them, and delivers the results in audio format. Notably, the smart glass hardware incorporates a built-in speaker for direct audio output and an earphone port for connecting to smartphones, enabling users to receive real-time audio feedback about recognized objects and their surroundings. The research specifically concentrated on object detection, color detection, face recognition, currency detection, and seamless integration of Speech-to-Text and Text-to Speech technologies, enhancing users' situational awareness through voice assistance.

2.1. Object detection:

In this research, the open-source MS COCO (Microsoft Common Objects in Context) dataset was utilized to develop the object detection model. This dataset encompasses 165,000 labeled images covering 90 distinct classes [9]. Remarkably, the MS COCO dataset provides annotations for various tasks, including object detection, captioning, keypoint detection, stuff image segmentation, panoptic, and dense pose. However, in this study, the dataset was exclusively employed for object detection purposes. Subsequently, the object detection model was constructed using the deep neural network-based YOLOv4 tiny (You Only Look Once version 4-tiny) Figure 2. The Yolov4-tiny method is designed based on the Yolov4 method to make it have a faster speed of object detection. The YOLOv4-tiny object detection achieves a remarkable speed of 443 frames per second with an RTX 2080Ti GPU, ensuring both speed and accuracy suitable for real-world applications. This advancement enhances the viability of deploying object detection methods on embedded systems or mobile devices. The

YOLOv4-tiny method employs the CSPDarknet53-tiny network as its backbone architecture. The CSPDarknet53-tiny network, the backbone architecture of YOLOv4-tiny, utilizes the CSPBlock module within the cross stage partial network instead of the ResBlock module in the residual network. The CSPBlock module divides the feature map into two parts and combines them using a cross stage residual edge. This approach enables gradient flow to propagate in two distinct network paths, enhancing the correlation difference of gradient information and thereby augmenting the learning ability of the convolution network when compared to the ResBlock module. Despite a 10%-20% increase in computation, this modification significantly improves accuracy. To optimize computation, computational bottlenecks with higher calculations in the CSPBlock module are removed, preserving or reducing the computational load. To simplify the computation process further, YOLOv4-tiny employs the LeakyReLU activation function in the CSPDarknet53-tiny network. In the feature fusion phase, YOLOv4-tiny employs a feature pyramid network to extract feature maps with varying scales, thereby increasing object detection speed. Unlike YOLOv4, it omits spatial pyramid pooling and path aggregation networks. YOLOv4-tiny utilizes feature maps of two different scales (13×13 and 26×26) to predict detection results, assuming an input image size of 416×416 and 80 feature classes for classification.

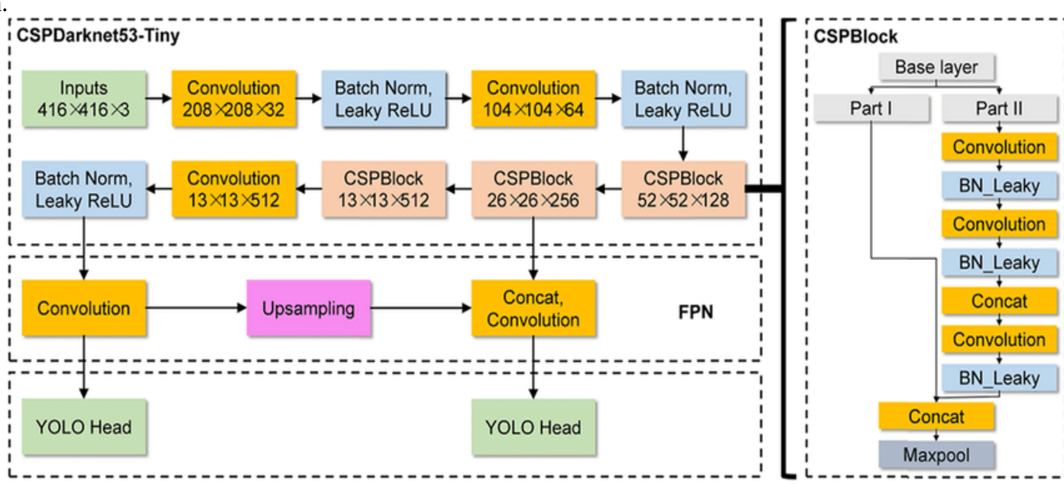


Figure 2. The architecture of YOLOv4-tiny Network [10]

The YOLOv4-tiny network structure, considering these modifications, is presented in Figure 2.

2.2. Face Recognition:

In the realm of assistive technologies, face recognition holds paramount significance for individuals with visual impairments, particularly the blind and visually impaired. This technology plays a pivotal role in enabling them to discern individuals, navigate complex social interactions, and uphold a semblance of autonomy. For individuals deprived of sight, identifying people through auditory cues or non-visual means poses considerable challenges. Face recognition technology emerges as a valuable supplementary resource, offering an additional layer of information and enhancing their overall sensory perception. Face recognition technology has undergone significant advancements due to the integration of deep learning models. This approach proves to be exceptionally accurate and robust, particularly when faced with challenges such as varying lighting conditions and facial occlusions like glasses or hats. However, the precision of face recognition systems heavily relies on the quality and quantity of data utilized for training these deep learning models. In this section, we present an in-depth exploration of a face recognition library that leverages a combination of innovative techniques, demonstrating its effectiveness in diverse scenarios.

2.2.1. Face Detection:

The first step in the face recognition process involves face detection. The library employs a pretrained face detection model, amalgamating HaarCascades [11], Histogram of Oriented Gradients (HOG) features [12], and deep neural networks. This hybrid model enables the identification of facial locations within an input image, even in the presence of partial occlusions, variations in angles, or distinct lighting conditions [13].

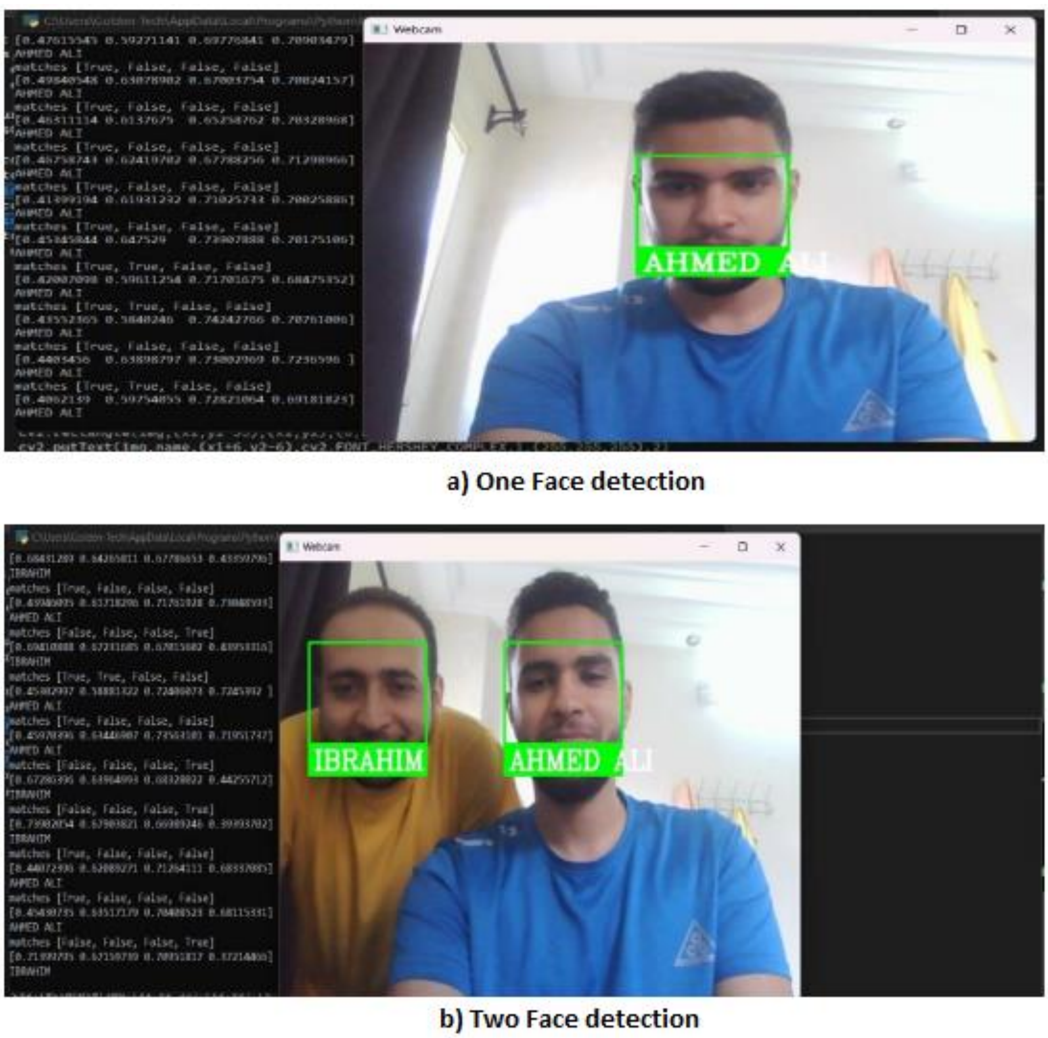


Figure 3. llustration of Face Detection Scenarios: (a) Single Face Detection, (b) Multi-Face Detection

2.2.2. Face Alignment

Following face detection, the library utilizes a technique known as ”face landmark estimation” [14] to pinpoint essential facial features, such as eyes, nose, and mouth. Subsequently, the face is aligned to a standardized pose. This alignment step is crucial for ensuring accurate recognition in subsequent stages [15].

2.2.3. Face Encoding

Upon alignment, the library extracts a numerical representation of the face, often referred to as a face embedding vector [16]. This vector encapsulates the unique facial features and is generated using a deep neural network trained on a vast dataset of faces. This encoding process captures intricate facial details, allowing for precise differentiation between individuals [17].

2.2.4. Face Recognition

To recognize a face, the library compares the extracted face encoding of the input image with a database of pre-stored face encodings. This comparison is performed using distance metrics such as Euclidean distance [18] or cosine similarity [19]. If the computed distance between the input face encoding and a stored face encoding falls below a predefined threshold, the library identifies the faces as a match, indicating successful recognition. The results obtained from the face recognition library are promising. The system accurately detects faces in input images, regardless of varying angles and lighting conditions. Each detected face is labeled with the corresponding person's name, and a bounding box is drawn around it, demonstrating the system's ability to provide precise and reliable face recognition outcomes. These results highlight the potential of deep learning models in advancing the field of face recognition technology, paving the way for enhanced applications in various domains.

In Figure 3, the image vividly demonstrates the remarkable versatility of the face recognition system. In subfigure (a), the system displays exceptional accuracy in detecting a single face, showcasing its precise capability in identifying individual subjects. Meanwhile, subfigure (b) exemplifies the system's advanced functionality as it adeptly recognizes and labels multiple faces, even in complex scenarios involving multiple individuals. This visual representation serves as compelling evidence, underscoring the system's proficiency in handling diverse face-detection scenarios with unparalleled precision.

Figure 4. Egyptian currencies [22]

2.3. Currency Detection

The development of currency detection technology for the visually impaired relies on a deep learning algorithm designed to recognize different denominations of Egyptian currency notes [20]. This innovative system captures input frames from a webcam and processes them to identify the specific denomination within the frame. Initially, the system loads a dataset of currency papers from Kaggle Figure 4, categorizing them into distinct classes to facilitate the training process. A pre trained MobileNetV2 architecture [21] is then utilized to train the model using images of Egyptian currency notes. The resulting trained model is preserved for future applications. Prior to feeding the images into the deep learning model, preprocessing steps, including resizing to a consistent format of (224, 224) pixels and the application of Gaussian blurring, are employed to enhance image quality. In this study, seven distinct output classes (1, 5, 10, 20, 50, 100, 200) are defined to represent the various denominations of Egyptian currency notes. The model architecture is meticulously crafted using the Keras Sequential API [23] following the MobileNetV2 base. It comprises four convolutional blocks, each incorporating multiple Conv2D layers, MaxPooling2D layers, and Dropout layers. Subsequent to the convolutional blocks, two dense layers are incorporated into the architecture, enabling advanced feature extraction and classification. During the model training phase, the Adam optimizer is employed, alongside Sparse Categorical Cross-Entropy as the loss function, and Accuracy as the evaluation metric. The model is meticulously compiled and trained using an extensive dataset, ensuring the network's ability to discern intricate patterns among different currency notes.

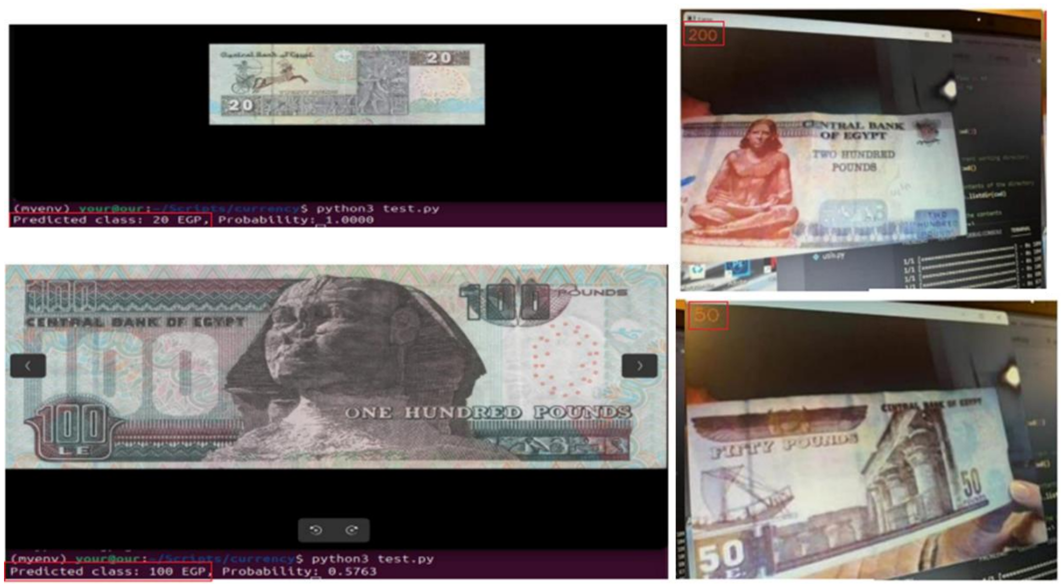


Figure 5. Predicted currency detection results, with the identified output highlighted within a red rectangle.

To assess the model's accuracy and reliability, a rigorous evaluation is conducted using an independent testing dataset results as shown in Figure 5. Following successful evaluation, the trained model is saved in the Hierarchical Data Format version 5 (HDF5) for future applications. In real-time currency detection scenarios, the system utilizes live-stream camera input. Each frame captured by the camera undergoes preprocessing, including resizing, normalization, and reshaping, aligning with the model's input requirements. The model predicts the currency class with the highest probability for each preprocessed frame, and the predicted class label is added to the frame, enabling real-time currency detection in practical and varied scenarios.

2.4. Color Detection

The identification of colors is essential for object recognition and finds applications in diverse areas such as image editing and drawing applications. This process involves determining the specific name associated with a given color.

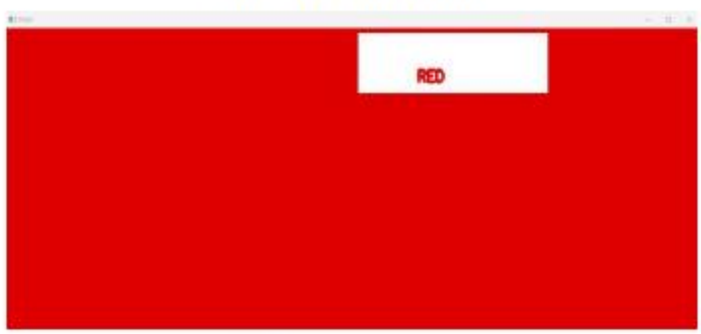
In this research, Python is utilized in three key components:

- Python Code for Color Recognition: This part involves the development of Python code to recognize colors within images.
- Image for Color Recognition Testing: An image is chosen to test the color recognition capabilities of the Python code.
- .csv File as Dataset: A .csv file acts as the dataset, containing various colors and their corresponding data.

These three modules collectively enable the research's objective: detecting colors within an image using Python. Initially, the image of interest was accessed and converted from the BGR color space to the HSV color space, allowing for a more robust analysis of color information. Subsequently, a circular region was defined at the center of the frame, and the HSV values within this area were extracted individually, separating the hue and saturation values. To ascertain color detection, the extracted HSV values were compared against predefined ranges. If the hue and saturation values fell within the specified range, the color was considered detected.

White color detection

Violet color detection



Red color detection

Figure 6. Color Detection Results

Conversely, if the values did not align with the defined range, the color was not detected. In the final step, a visual representation of the detected color was generated by drawing a circle around the specified area see Figure 6. Additionally, textual information indicating the detected color was overlaid on the image for further analysis and interpretation. In addition to the visual representation, upon detecting a specific color, the system was designed to transmit the information to headphones, providing an auditory feedback loop to the user.

2.5. Speech to Text and Text to Speech

2.5.1. Speech to Text

In this scientific investigation, a sophisticated speech-to-text functionality was developed through the integration of various specialized libraries. The ’speech recognition’ library played a central role, providing an efficient interface for capturing audio input and converting it into textual form. This allowed spoken words and phrases to be accurately transcribed, enabling systematic processing and analysis of the spoken content.

To enhance the functionality of the speech-to-text system, the ’translatepy’ library was seamlessly integrated. This addition enabled the translation of recognized text into multiple languages, significantly improving the accessibility and usability of the application. Particularly beneficial in multilingual contexts, this feature addressed diverse language content scenarios.

Challenges associated with languages featuring complex scripts, such as Arabic, were effectively managed using the ’bidi.algorithm’ and’ arabic reshaper’ libraries. These resources were instrumental in overcoming issues related to right to-left text orientation and intricate Arabic script shaping. Through reshaping and alignment techniques, the readability and visual presentation of transcribed Arabic content were notably improved.

Furthermore, the incorporation of the ’keyboard’ library facilitated the integration of intuitive keyboard shortcuts and hotkeys within the application. This functionality empowered users to seamlessly execute various actions, including initiating or halting the speech recognition process, switching between languages, and activating specific functions within the system.

The implemented system operates in a structured manner. Users are prompted to specify their preferred language, after which they speak into a connected microphone. The ’speech recognition’ library captures and records the speech, further refined by Google’s speech recognition API for precise text recognition. Post-recognition, the text undergoes reshaping and formatting if necessary. Subsequently, translation into English is performed using the ’translatepy’ library. The resulting translated text then serves as input commands, directing subsequent actions within the system.”

2.5.2. Text to Speech

In this research, the EAST (Efficient and Accurate Scene Text) [24] model is utilized for text detection, complemented by Tesseract OCR (Optical Character Recognition) [25] for text recognition. Real-time input is sourced from the camera feed. The EAST model, built on a fully convolutional neural network, excels at accurately identifying text in diverse orientations and sizes. Trained on the

SynthText dataset, comprising over 800,000 images with synthetic text, the model has undergone rigorous evaluation across multiple text detection benchmarks.

The code initiates by loading the EAST text detector model. Following this, it initializes the camera object and enters a loop to continuously capture frames from the camera feed. These frames are resized to 640x320 pixels, converted to grayscale, and subjected to adaptive thresholding. Blob frames are then created from these frames and passed through the EAST model for text detection. To eliminate overlapping bounding boxes, non-maximum suppression is applied. Text recognition within each bounding box is achieved using Tesseract OCR, a robust open-source OCR engine proficient in recognizing over 100 languages.

In the final output, the code delineates bounding boxes around each detected text region and displays the resulting frame. The program actively monitors for a key press to exit the loop. Upon exiting, the camera object is released, and all display windows are closed. In this research, we implemented a text-to-speech functionality using a combination of specialized libraries. A fundamental component of the approach involved integrating the ’langdetect’ library, enabling automatic language detection of the input text. This capability was instrumental in selecting appropriate pronunciation and accent settings for the subsequent text-to speech conversion process.

For the conversion of text to speech, we employed the ’gtts’ (Google Text-to Speech) library. This library provided a convenient interface to harness the robust text-to-speech capabilities offered by Google’s TTS engine. Bypassing the desired text to the ’gtts’ library, we generated corresponding audio files in multiple languages and voice options.

To facilitate audio playback, we utilized the ’pygame’ library, which offers a versatile set of functions for sound and music management. Integrating ’pygame’ into the research enabled seamless playback of the generated audio files, allowing control over playback parameters such as volume, speed, and looping.

To optimize user interaction, we leveraged the ’os’ library, which facilitated communication with the underlying operating system. This interaction allowed efficient management of files and directories, ensuring effective storage and retrieval of the generated audio files.

By orchestrating the functionalities of these libraries, a text-to-speech feature was successfully implemented within the research. Users can input text, and the application seamlessly converts it into high-quality speech in their preferred language and voice. This functionality significantly enhances accessibility and offers an alternative mode of communication, thereby augmenting the user friendliness and versatility of the research.

The text-to-speech process involves several key steps. Initially, the input text is acquired as a string from Optical Character Recognition (OCR). Subsequently, the ’langdetect’ library is employed to automatically detect the language of the input text. Following language identification, the ’gtts’ library is utilized to transform the text into an audio file. The resulting audio file is then played back to the user using the ’pygame’ library. Finally, the ’os’ library is utilized to manage the deletion of the temporary audio file, completing the text-to-speech process.

3. Hardware Components

The ultimate goal of this work is to design a novel navigation-aiding device for object identification and atmospheric environment description. A list of hardware components used to implement this concept is illustrated in Figure 7, which is explained below:

- **Raspberry Pi 4 Model B+:** This primary device is chosen for its cost-efficiency, portability, and efficient performance compared to other embedded systems.
- **ESP32-CAM Development Board (with Camera):** The ESP32 Camera module includes an OV2640 camera sensor, capable of capturing 2-megapixel photos and 720p videos. It can be programmed using various platforms such as the Arduino IDE, MicroPython, and the Espressif IoT Development Framework (ESP-IDF). Connect the ESP32 Camera module to the Raspberry Pi using jumper wires. The connection pins on the ESP32 Camera module should be connected to the corresponding GPIO pins on the Raspberry Pi.

- **Ultrasonic Sensor:** This sensor provides distance measurement, allowing the device to detect obstacles and navigate effectively within the environment. An ultrasonic sensor is connected to a Raspberry Pi using the GPIO pins (connected to GPIO pins 17 (TRIG) and 18 (ECHO)).
- **Vibration Motor:** A vibration motor is integrated into the design to provide haptic feedback, enhancing user interaction and providing alerts or notifications as needed. A Vibration Motor is connected to a Raspberry Pi using the GPIO pins (connected to GPIO pins 13)

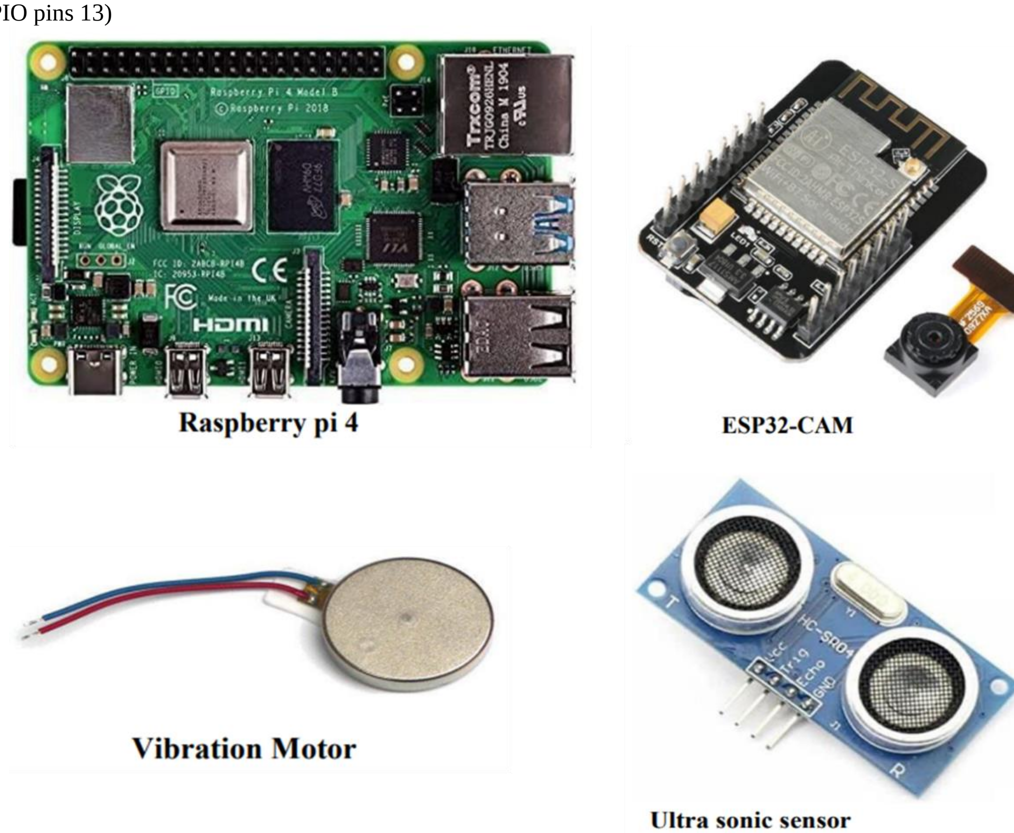


Figure 7. Hardware Components of Smart Glasses

These carefully selected hardware components form the foundation of the novel navigationassisting device, ensuring efficient object identification and description of the surrounding atmosphere. The utilization of the Raspberry Pi 4 Model B+ as the core device, in combination with the ESP32-CAM Development Board, ultrasonic sensor, and vibration motor, enables the creation of a functional and user-friendly navigation aid.

Discussion of the different stages of the algorithm 1

Step 1-5: The initial steps involve setting up the hardware and software components. The Raspberry Pi 4 is initialized and essential object detection models (YOLOv4 Tiny, MobileNetV2, and HaarCascade) are loaded onto the system. The system's hyperparameters, crucial for the models' performance, are optimized at this stage. Additionally, the ESP32 CAM camera is configured to stream video frames, and the audio output device is prepared to provide feedback. These steps are foundational, establishing the necessary infrastructure for the object detection process.

Step 6: This step constitutes the core of the object detection process and is iteratively applied to each video frame captured by the Raspberry Pi camera.

Step 6.1: The frame is passed through the YOLOv4 Tiny object detection model, which identifies objects within the frame. YOLOv4 Tiny is particularly efficient for real-time object detection tasks due to its optimized architecture.

Step 6.2: The frame is processed through two additional models: the MobileNetV2-based Currency Detection model and the HaarCascade-based Face Detection model. MobileNetV2 is a lightweight deep-learning model suitable for tasks involving mobile and edge devices, making it ideal for currency detection. HaarCascade is a feature-based object detection method often used for face detection due to its accuracy and speed.

Step 6.3-6.4: After detecting objects, including currency denominations, colors, and recognized faces, the system generates an audio feedback message specifying the detected object class, currency denomination, and recognized faces. This audio feedback is then played through headphones, providing real-time auditory information to the user.

Step 7: The entire process in Step 6 is repeated for every subsequent frame in the video stream. The iterative nature of this step ensures that the system continuously analyzes the real-time video feed, providing continuous feedback to the user.

Step 8: Finally, the system is designed to handle interruptions or the cessation of the video stream gracefully. If the video stream stops or the program is interrupted, all resources utilized by the system are released, ensuring efficient operation and resource management.

Algorithm: Proposed System

1. **Step 1:** Initialize the Raspberry Pi 4 and load the required object detection models.
2. **Step 2:** Optimize the hyperparameters of YOLOv4 Tiny, MobileNetV2, and HaarCascade models.
3. **Step 3:** Initialize the ESP32-CAM camera and configure it to stream video frames.
4. **Step 4:** Set up the audio output device to provide feedback.
5. **Step 5:** Begin streaming video frames from the Raspberry Pi camera.
6. **Step 6:** For each video frame:
   - **Step 6.1:** Pass the frame through the Object Detection model (YOLOv4 Tiny).
   - **Step 6.2:** Process the frame through the MobileNetV2-based Currency Detection model and HaarCascade-based Face Detection model.
   - **Step 6.3:** Generate an audio feedback message indicating the detected object class, currency denomination, and recognized faces.
   - **Step 6.4:** Play the generated audio feedback through the headphones.
7. **Step 7:** Repeat Step 6 for every subsequent frame in the video stream.
8. **Step 8:** If the video stream stops or the program is interrupted, release all resources used by the system.

4. Discussion

The proposed system represents a significant advancement in assistive technology for visually impaired individuals, integrating various components to enhance independence and navigation. The system's utilization of computer vision techniques, including object detection, color detection, and face recognition, provides users with valuable information about their surroundings. By leveraging deep learning algorithms such as YOLOv4-tiny and MobileNetV2, the system achieves real-time object and currency detection with high accuracy. Furthermore, the inclusion of HaarCascades for face detection ensures reliable recognition of individuals, contributing to enhanced social interaction and situational awareness. One of the system's strengths lies in its seamless integration of speech-to-text and text-to-speech functionalities. By converting spoken words into text and providing audible feedback, the system enables natural interaction and communication for visually impaired users. Moreover, the ability to translate text into multiple languages enhances accessibility and usability in diverse linguistic contexts.

The hardware components used in the system, including the Raspberry Pi 4 Model B+, ESP32- CAM Development Board, and ultrasonic sensor, form a robust foundation for navigation assistance.

These components offer a balance of cost-effectiveness, portability, and performance, making the device suitable for real-world applications.

In terms of future improvements, there are several avenues for enhancement. Further optimization of deep learning models could improve accuracy and expand the range of objects and facial features recognized by the system. Additionally, incorporating machine learning techniques to adapt to different environmental conditions, such as varying lighting or background noise, would enhance the system's reliability in diverse settings.

User feedback and usability studies are essential for refining the system to better meet the needs of visually impaired individuals. Collaborations with user communities and organizations can provide valuable insights into usability challenges and preferences, guiding iterative improvements to the device's design and functionality.

Exploring opportunities for miniaturization and wearable solutions could enhance the device's portability and integration into users' daily lives. Wearable devices, such as smart glasses, could offer a hands-free navigation experience, further enhancing user independence and mobility.

Overall, the proposed system represents a promising step towards empowering visually impaired individuals and improving their quality of life. By combining advanced technology with user-centered design principles, the system addresses crucial challenges faced by the visually impaired community and offers a pathway to greater independence and accessibility.

5. Conclusion

The paper presents a comprehensive study on the development of an electronic device aimed at enhancing the independence of visually impaired individuals. The device incorporates object detection and audible feedback to assist in navigation and increase environmental awareness for visually impaired users. The research is motivated by the significant global population (2.2 billion individuals) dealing with visual impairments, highlighting the urgency of addressing this issue. Several related works in the field of assistive technologies are discussed, showcasing the progress made in employing intelligent systems to aid visually impaired individuals.

The proposed system integrates various components, including object and color detection, face recognition, currency detection, speech-to-text, and text-to-speech functionalities. The system utilizes a Raspberry Pi 4 Model B+ as the central processing unit and an ESP32-CAM Development Board with a camera for capturing live video feeds. The object detection module employs the YOLOv4 tiny algorithm, known for its speed and accuracy, to identify objects in real time. The face recognition module combines HaarCascades, Histogram of Oriented Gradients (HOG) features, and deep neural networks for robust facial recognition. Additionally, a currency detection module is implemented using a MobileNetV2-based deep learning model to recognize Egyptian currency denominations.

The system also includes speech-to-text and text-to-speech functionalities. Speech-to-text conversion is achieved using the ’speech recognition’ library, and the translated text is processed and presented to the user. Text-to-speech conversion utilizes the ’gtts’ (Google Text-to-Speech) library, offering high-quality speech synthesis. The system seamlessly integrates these functionalities, enabling users to interact with the device effectively.

Furthermore, the paper discusses the hardware components used in the implementation, including the Raspberry Pi 4 Model B+, ESP32-CAM Development Board with camera, ultrasonic sensor, and vibration motor. These components form the basis of the navigation-aiding device, facilitating object identification and providing feedback to users.

The proposed system combines advanced computer vision techniques, deep learning algorithms, and effective hardware components to create a versatile assistive device for visually impaired individuals.

By integrating object and color detection, face recognition, currency detection, speech-to-text, and text-to-speech functionalities, the system aims to enhance the independence and mobility of visually impaired users, providing real-time information about their surroundings and aiding navigation.

The paper presents a holistic approach, addressing various aspects of assistive technologies, and demonstrates the potential of intelligent systems in improving the lives of visually impaired individuals.

In the realm of future work, the system’s effectiveness could be further enhanced through continuous refinement of its components. Exploring advanced deep learning architectures for object detection and face recognition might elevate accuracy and expand the system’s capability to recognize a broader range of objects and facial features. Additionally, incorporating machine learning techniques to dynamically adapt to different environmental conditions could bolster the system’s performance. Furthermore, integrating user feedback mechanisms and conducting usability studies would provide valuable insights, allowing for user-centered improvements. Collaborations with visually impaired individuals and organizations can offer unique perspectives, guiding the system’s evolution to better cater to the specific needs and preferences of its users. Lastly, exploring opportunities for miniaturization and wearable solutions could make the device more portable and seamlessly integrated into users’ daily lives, ensuring greater accessibility and independence.

6. References

[1] Steinmetz, John D., Ronald R. Bourne, Paul S. Briant, Steven R. Flaxman, Hugh R. Taylor, Jost B. Jonas, Abdol A. Abdoli, Wubetu A. Abrha, Ayman Abualhasan, and Esraa G. Abu-Gharbieh. "Causes of blindness and vision impairment in 2020 and trends over 30 years, and prevalence of avoidable blindness in relation to Vision 2020: The right to sight: An analysis for the Global Burden of Disease Study." The Lancet Global Health 9, no. 2 (2021): 144–160.

[2] Daescu, Oana, Hongwei Huang, and Martin Weinzierl. "Deep learning-based face recognition system with smart glasses." In Proceedings of the 12th ACM International Conference on PErvasive Technologies Related to Assistive Environments, 218–226, 2019.

[3] Joshi, Ravi C., Shashank Yadav, Monojit K. Dutta, and Carlos M. Travieso-Gonzalez. "Efficient multi-object detection and smart navigation using artificial intelligence for visually impaired people." Entropy 22, no. 9 (2020): 941.

[4] Awad, Muhammad, Jad El Haddad, Elie Khneisser, Tamara Mahmoud, Elias Yaacoub, and Michel Malli. "Intelligent eye: A mobile application for assisting blind people." In 2018 IEEE Middle East and North Africa Communications Conference (MENACOMM), 1–6, 2018. IEEE.

[5] Bhatlawande, Siddharth, Shilaskar Shrirang, Abhyankar Abhijit, Ahire Mayur, Chadgal Aniket, and Madake Jay. "Detection of exercise and cooking scene for assistance of visually impaired people." In Pervasive Computing and Social Networking: Proceedings of ICPCSN 2022, 493–508, Springer, 2022.

[6] Mukhiddinov, Muzaffar, Akmal Muminov, and Jungpil Cho. "Improved classification approach for fruits and vegetables freshness based on deep learning." Sensors 22, no. 21 (2022): 8192.

[7] Khan, Mohammad A., Pallab Paul, Md Rashid, Md Hossain, and Md A. R. Ahad. "An AI-based visual aid with integrated reading assistant for the completely blind." IEEE Transactions on Human-Machine Systems 50, no. 6 (2020): 507–517.

[8] Nishajith, A., J. Nivedha, S. S. Nair, and J. M. Shaffi. "Smart cap-wearable visual guidance system for blind." In 2018 International Conference on Inventive Research in Computing Applications (ICIRCA), 275–278, 2018. IEEE

[9] Lin, Tsung-Yi, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. "Microsoft COCO: Common objects in context." In Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13, 740–755, Springer, 2014

[10] Saponara, Sergio, Ahmed Elhanashi, and Qiang Zheng. "Developing a real-time social distancing detection system based on YOLOv4-tiny and bird-eye view for COVID-19." Journal of Real-Time Image Processing 19, no. 3 (2022): 551–563

[11] Mantoro, Teddy, Marina A. Ayu, et al. "Multi-faces recognition process using Haar cascades and Eigenface methods." In 2018 6th International Conference on Multimedia Computing and Systems (ICMCS), 1–5, 2018. IEEE.

[12] Déniz, Oscar, Guillermo Bueno, Juan Salido, and Fernando De La Torre. "Face recognition using histograms of oriented gradients." Pattern Recognition Letters 32, no. 12 (2011): 1598–1603.

[13] Rahmad, C., R. A. Asmara, D. Putra, I. Dharma, H. Darmono, and I. Muhiqqin. "Comparison of Viola-Jones Haar cascade classifier and histogram of oriented gradients (HOG) for face detection." In *IOP Conference Series: Materials Science and Engineering*, vol. 732, 012038, 2020. IOP Publishing.

[14] Zhu, Xiangxin, and Deva Ramanan. "Face detection, pose estimation, and landmark localization in the wild." In 2012 IEEE Conference on Computer Vision and Pattern Recognition, 2879–2886, 2012. IEEE.

[15] Wu, Y., and Q. Ji. "Facial landmark detection: A literature survey." International Journal of Computer Vision 127 (2019): 115–142.

[16] Shi, Yichun, and Anil K. Jain. "Probabilistic face embeddings." In Proceedings of the IEEE/CVF International Conference on Computer Vision, 6902–6911, 2019.

[17] Abdullah, Ibrahim A., et al. "Face recognition using face embedding method." Turkish Journal of Computer and Mathematics Education (TURCOMAT) 12, no. 10 (2021): 3383–3394.

[18] Wu, H., Y. Cao, H. Wei, and Z. Tian. "Face recognition based on Haar-like and Euclidean distance." In Journal of Physics: Conference Series, vol. 1813, 012036, 2021. IOP Publishing.

[19] Nguyen, Hieu V., and Li Bai. "Cosine similarity metric learning for face verification." In Asian Conference on Computer Vision, 709–720, Springer, 2010.

[20] Semary, Nagwa A., Samar M. Fadl, Mostafa S. Essa, and Ahmed F. Gad. "Currency recognition system for visually impaired: Egyptian banknote as a study case." In 2015 5th International Conference on Information & Communication Technology and Accessibility (ICTA), 1–6, 2015. IEEE.

[21] Bhadani, Abhishek K., and Abhishek Sinha. "A facemask detector using machine learning and image processing techniques." Engineering Science and Technology, an International Journal (2020): 1–8.

[22] Yousry, Amr, Mohamed Taha, and Mohamed M. Selim. "Currency recognition system for blind people using ORB algorithm." Int. Arab. J. e Technol. 5, no. 1 (2018): 34–40.

[23] Manaswi, Neelesh K., and Neelesh K. Manaswi. Understanding and Working with Keras. Deep Learning with Applications Using Python: Chatbots and Face, Object, and Speech Recognition with TensorFlow and Keras, 31–43, 2018.

[24] Zhou, X., C. Yao, H. Wen, Y. Wang, S. Zhou, W. He, and J. Liang. "EAST: An efficient and accurate scene text detector." In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 5551–5560, 2017.

[25] Mithe, R., S. Indalkar, and N. Divekar. "Optical character recognition." International Journal of Recent Technology and Engineering (IJRTE) 2, no. 1 (2013):